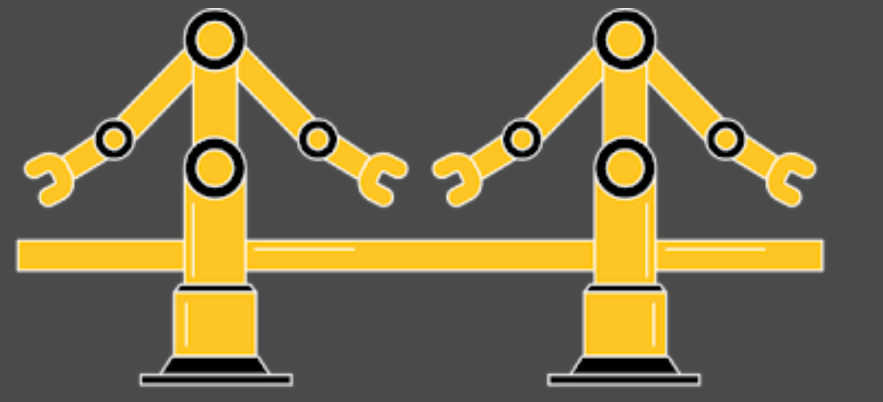


Beyond the Snowflakes: Disentangling Weather Effects from Confounding Meta-Factors in LiDAR Object Detection.



Michigan Tech

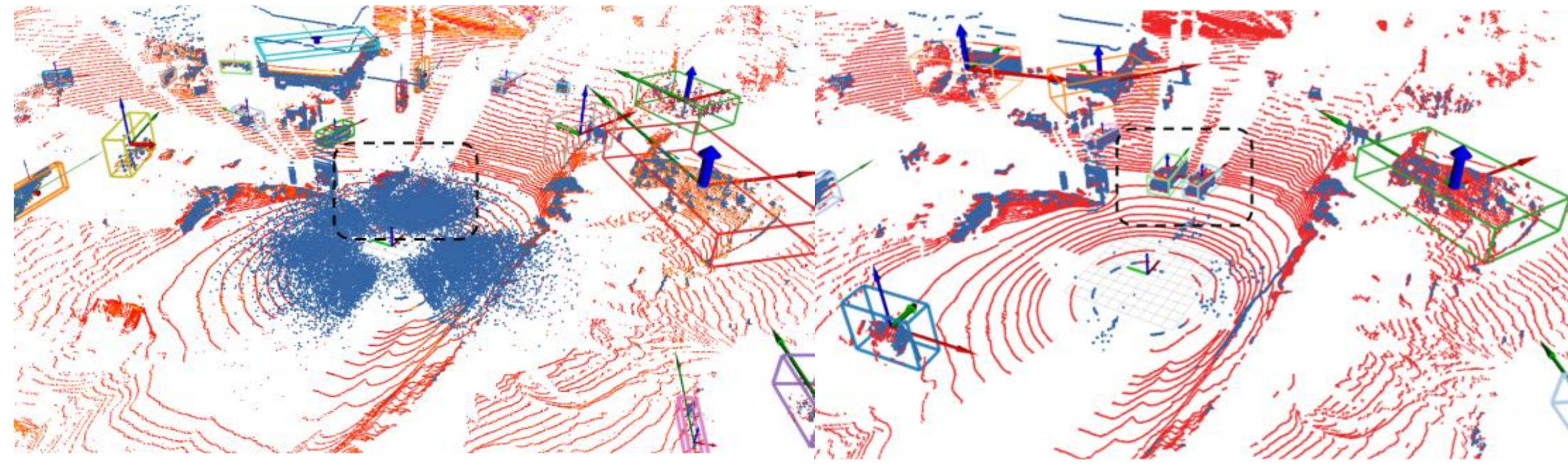


IROS 2026
PITTSBURGH

Yiming Yang and Jeremy P. Bos
Michigan Technological University

Research Question

How does snow influence LiDAR-based 3D object detection?



Existing snow simulator

- Only airborne snow
- No accumulated snow

Existing real-world evaluation

- Cross-domain variation
- Coarse categories (light vs. heavy snow)

Proposed Approach

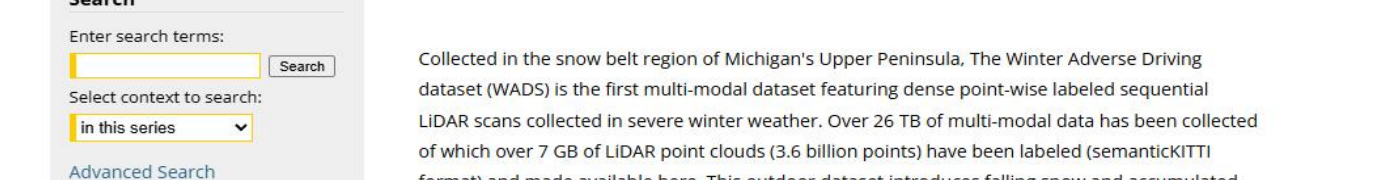
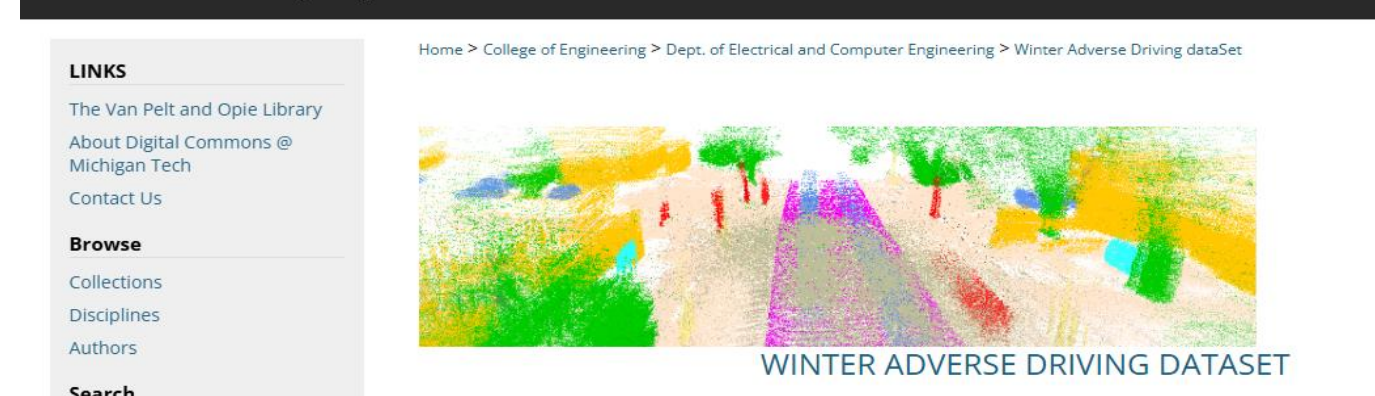
1. Granular snow metric - number of raw snow points

2. Diverse neural architectures

Detector	Architecture	Stages
SECOND-IoU	Voxel-based	One
Voxel R-CNN	Voxel-based	Two
PointPillar	Pillar-base	One
SECOND	Point-based	Two
PV-RCNN	Hybrid	Two

[OpenPCDet, 2020]

WADS & WADS-3D

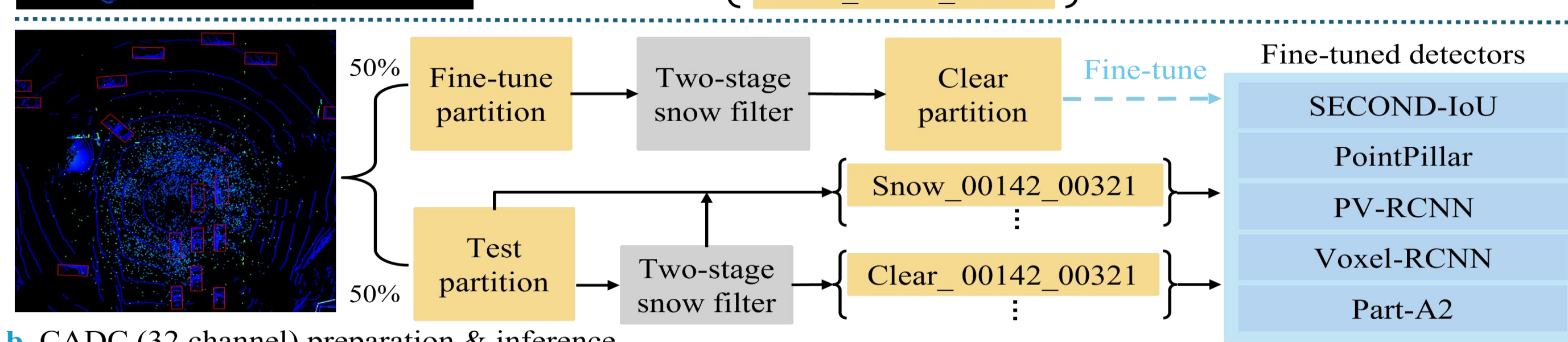
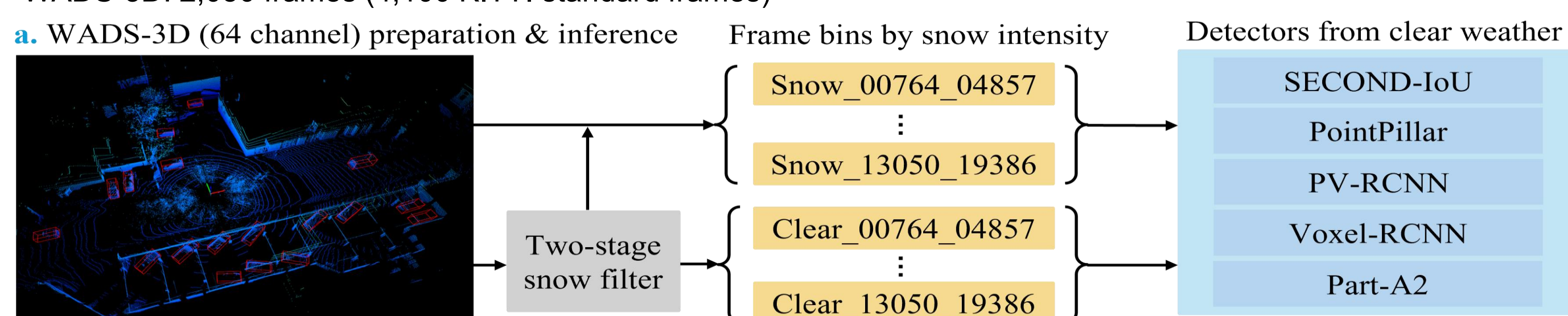


LiDAR | RGB | Thermal | GNSS
> 45 TB raw data
> 7GB semantic annotation
> 80K downloads

30,104 labeled cars
High-clutter snowy environments

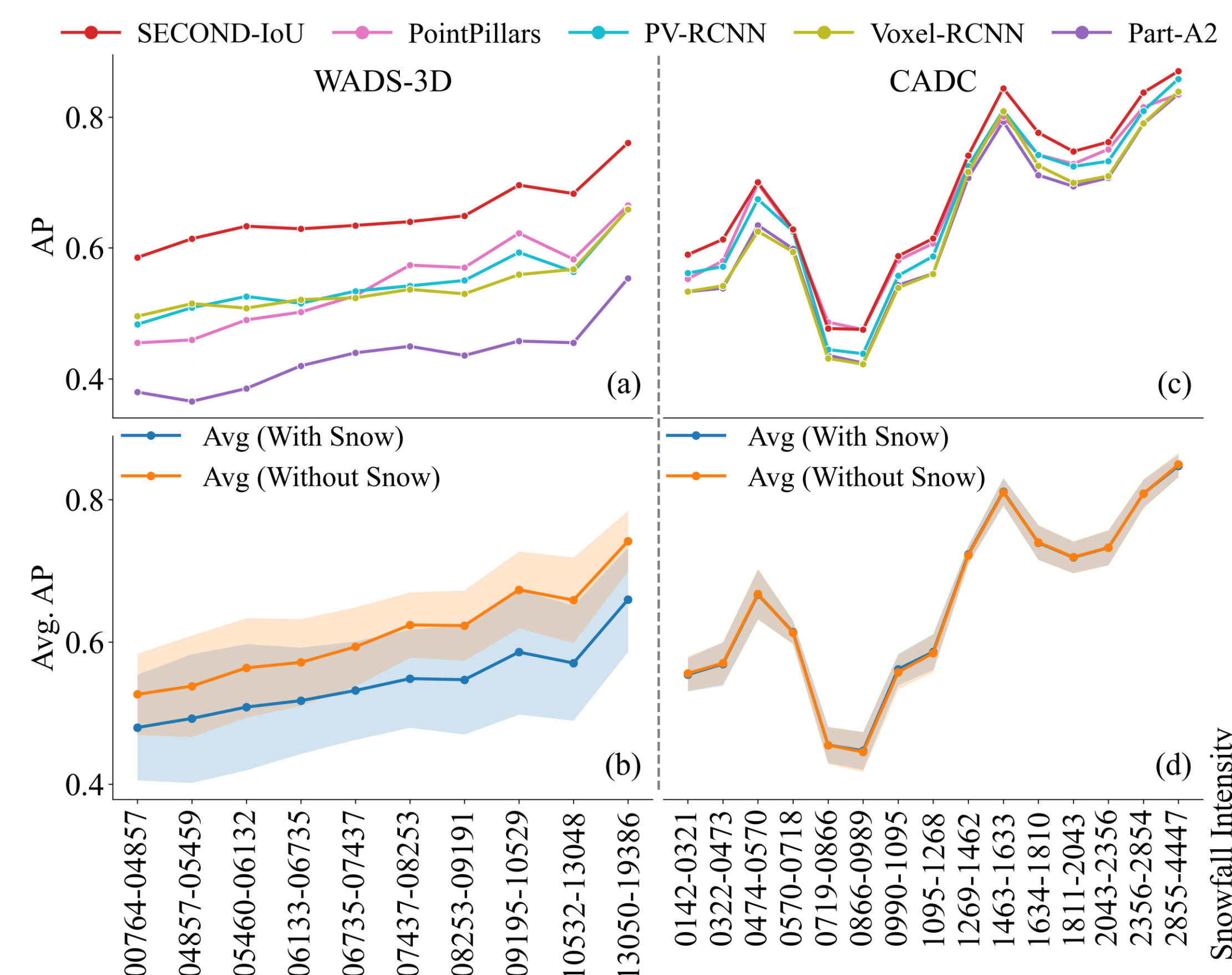
Evaluation Pipeline

WADS-3D: 2,050 frames (4,100 KITTI standard frames)

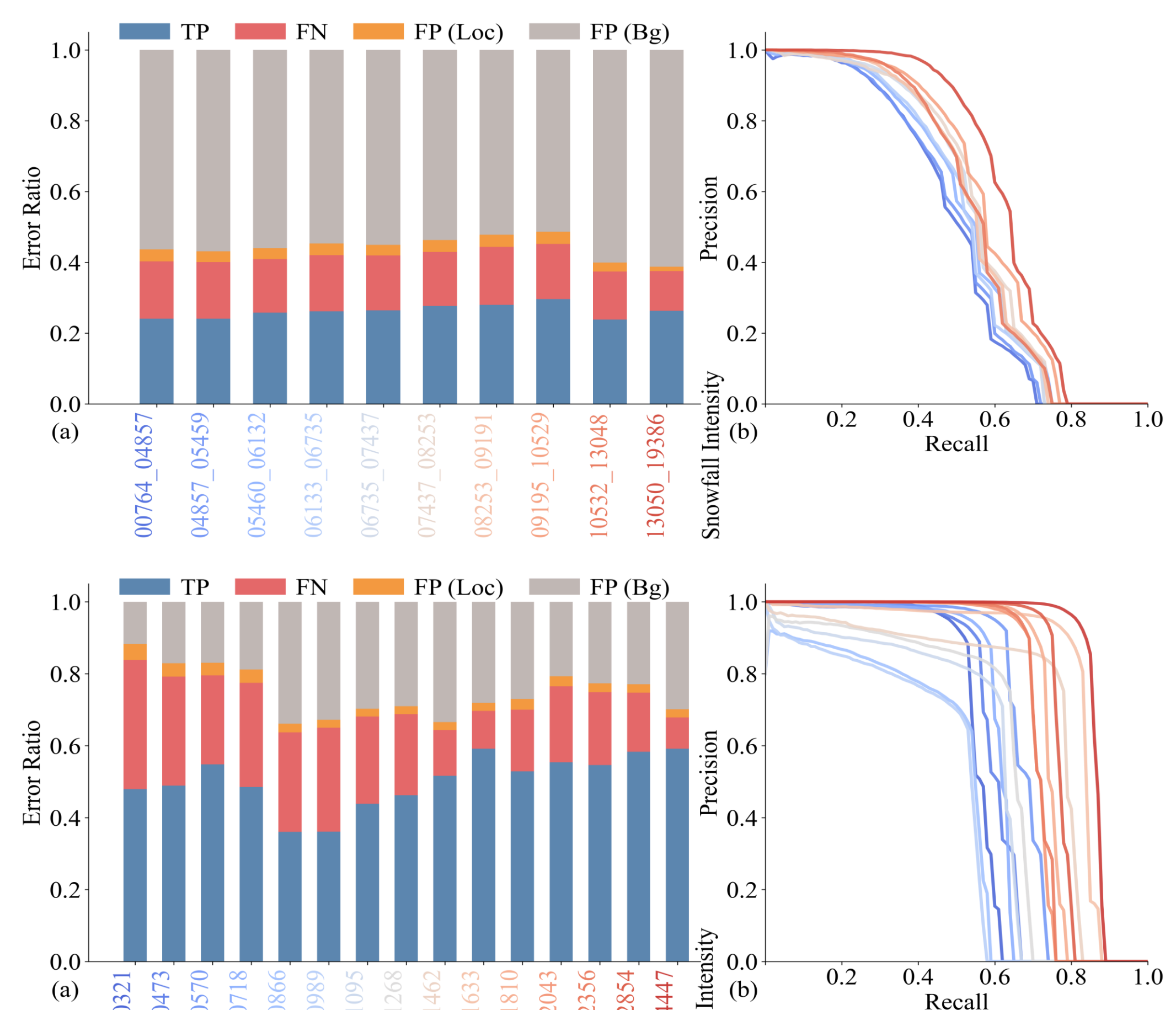


CADC: 7,000 frames (14,000 KITTI standard frames)

Overall Results & Ablation Study



Analysis: Error Matrix & Recall



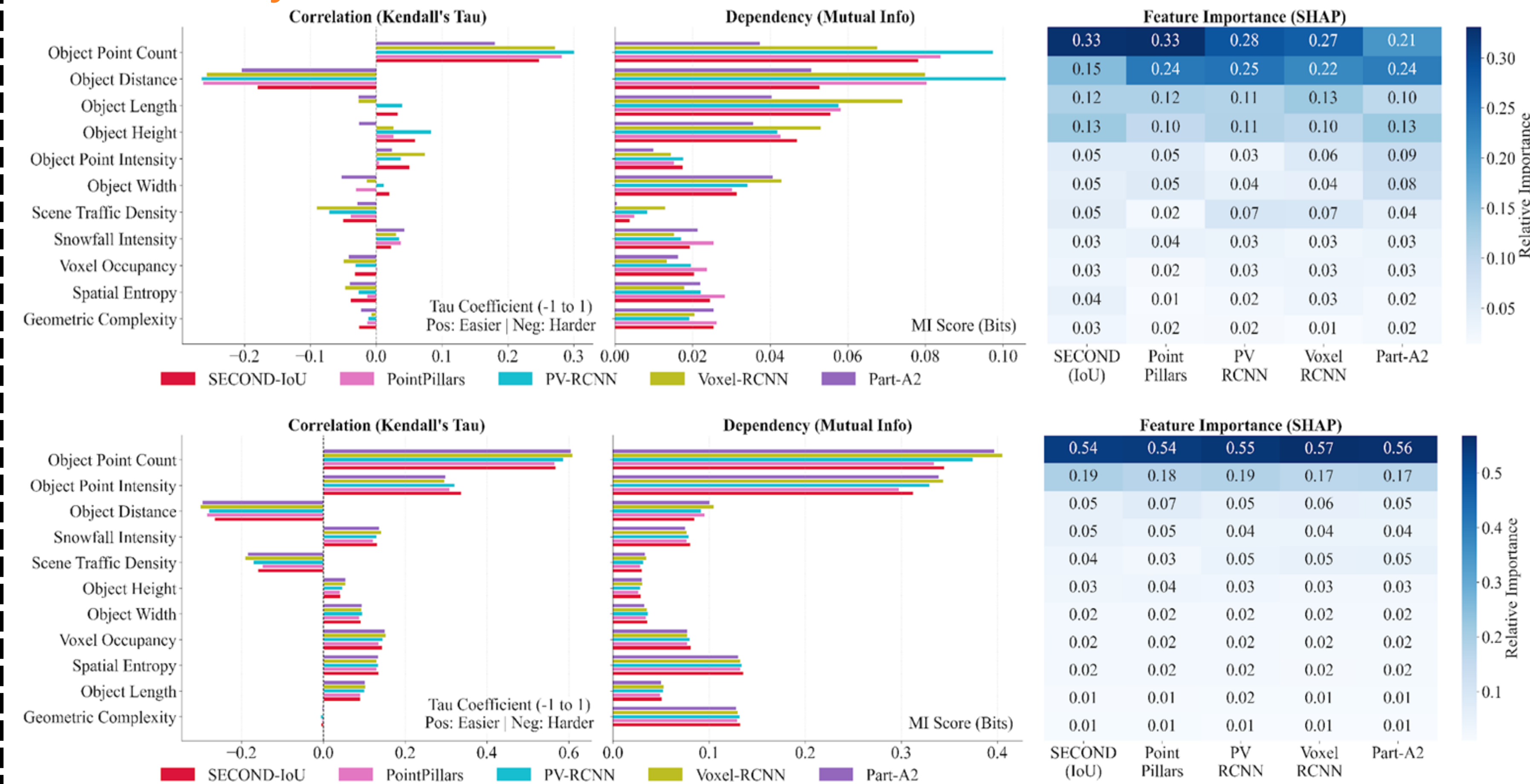
Hypotheses

Scene Smoother: Snow covers background clutter (vegetation, irregular road edges), suppressing environmental noise to make vehicle stand out.

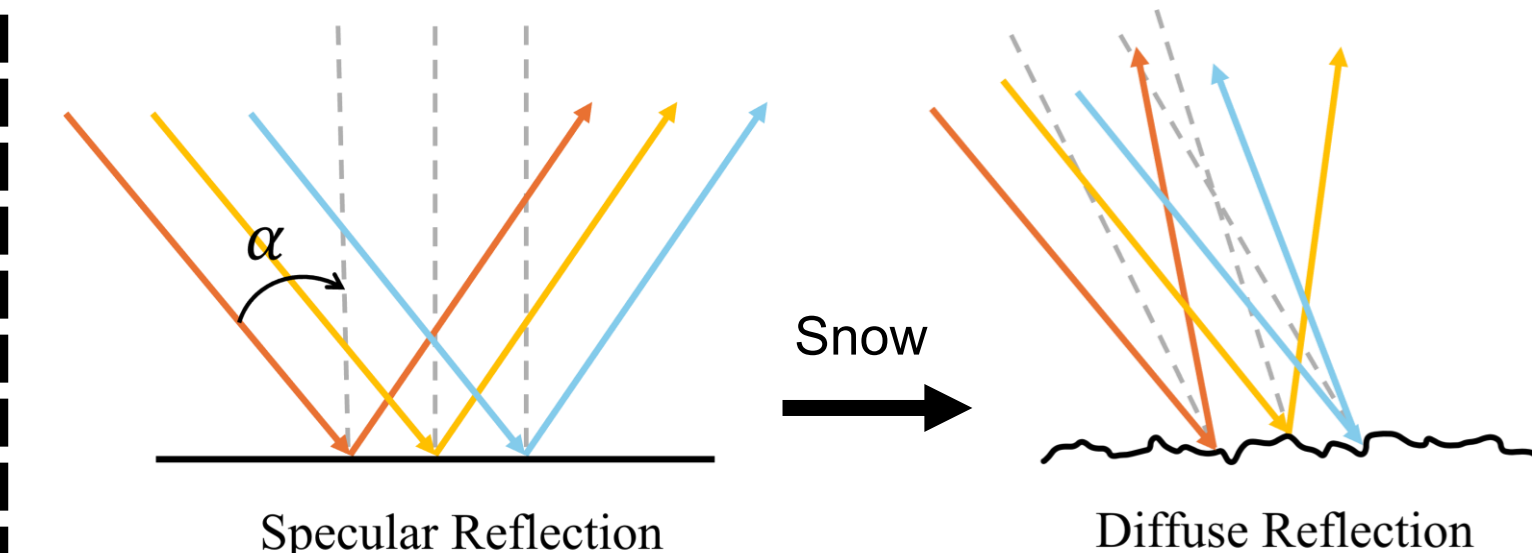
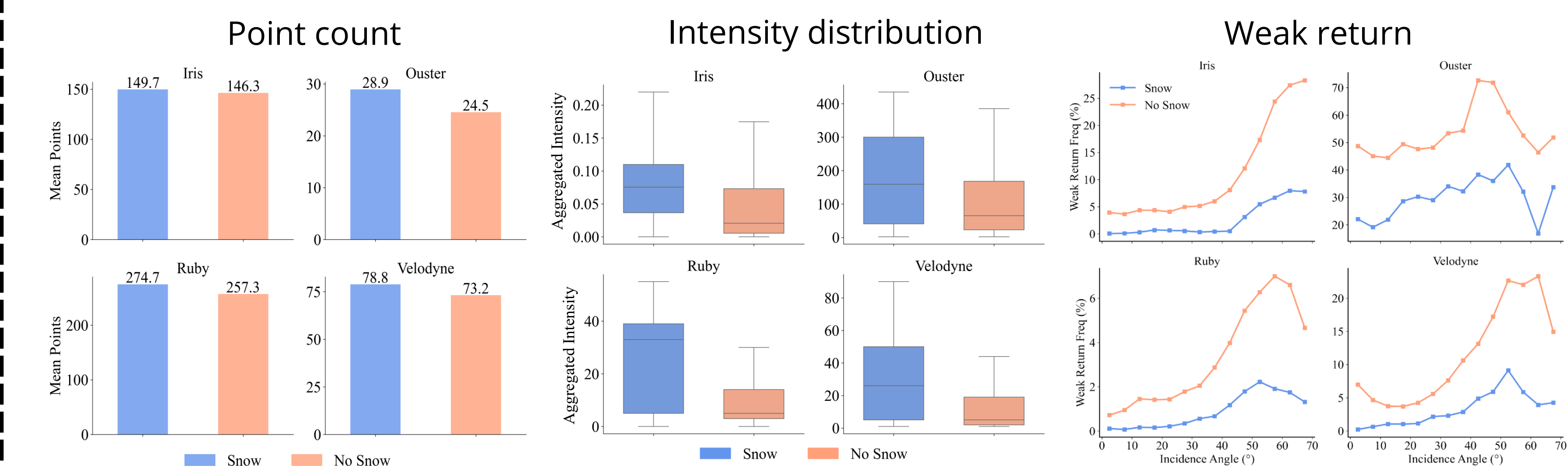
Geometric Enhancer: Falling snowflakes within the vehicle bounding box create extra lidar returns, artificially inflating point counts and boosting detectability.

Reflectance Transformer: Snow transforms specular (mirror-like) car paint into diffuse lambertian surface, increase return counts and signal intensity.

Meta-analysis



Controlled Experiment



- Accumulated snow boosts vehicle dimensions.
- Accumulated snow increases the reflection return point intensity.
- Accumulated snow makes the vehicle surface more diffusely reflective.

Future Work

- Find the cutoff point where heavy snow buildup distorts an object's shape and causes detection to fail.
- Test how snow affects LiDAR performance for other objects, such as pedestrians and cyclists.
- Study how different snow conditions, such as moisture, age, and texture, change how LiDAR signals bounce back in various winter environments.